Figure 1: Pract. Assignment

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.3     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(haven)


#Step 1: Upload data
PartisanVote <- read_dta(file.path("~/Desktop","cumulative_2006-2024.dta"), col_select = c(pi
  filter(year >= 2006 & year <= 2024)
```

```
#Step 2: recode partisan identity (pid7) "partisan identity".

#recode liberal variables: "Strong Democrat" = 1". "Not Very Strong Democrat" = 2.
#recode moderate variables: "Lean Democrat" = 3". "Independent" = 4. "Lean Republican" = 5.
#recode conservatives variables: "Not Very Strong Republican" = 6. "Strong Republican" = 7.
```

```r
class(PartisanVote$pid7)
```

```
[1] "haven_labelled" "vctrs_vctr"     "double"
```

```r
str(PartisanVote$pid7)
```

```
 dbl+lbl [1:701955] 1, 3, 1, 1, 1, 3, 7, 1, 1, 1, 7, 5, 5, 3, 4, 4, 3, 8, 3...
 @ label       : chr "Partisan identity (7 point)"
 @ format.stata: chr "%12.0g"
 @ labels      : Named num [1:9] 1 2 3 4 5 6 7 8 9
  ..- attr(*, "names")= chr [1:9] "Strong Democrat" "Not Very Strong Democrat" "Lean Democrat
```

```r
table(as.numeric(PartisanVote$pid7))
```

```
     1      2      3      4      5      6      7      8      9
173790  83093  70754  97588  69137  65183 118558  20817      3
```

```r
# Step 3: Create the 4 groups, tagged by panel
PartisanVote_Grouped <- PartisanVote %>%
  mutate(
    Grouped = case_when(
      pid7 %in% c(1, 7) ~ "Strong Identifiers",
      pid7 %in% c(2, 6) ~ "Weak Identifiers",
      pid7 %in% c(3, 5) ~ "Independent Leaners",
      pid7 == 4         ~ "Pure Independents"),
    panel = case_when(
      pid7 %in% c(1, 2, 6, 7) ~ "Identifiers",
      pid7 %in% c(3, 4, 5)    ~ "Leaners and Independents"))

# Step 4: Aggregate (one pass, all valid respondents in the denominator)
PartisanVote_agg <- PartisanVote_Grouped %>%
  filter(!is.na(Grouped)) %>%
  count(year, Grouped, panel) %>%
  group_by(year) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup()
```

```r
ggplot(data = PartisanVote_agg, mapping = aes(x = year, y = prop, color = Grouped)) +
  geom_point(size = 1.2, fill = "white", stroke = 0.5) +
  geom_line(linewidth = 0.7) +
  facet_wrap(~ panel, ncol = 1) +
  theme_bw(base_size = 6) +
  scale_y_continuous(limits = c(0, 0.5)) +
  scale_colour_manual(values = c(
    "Strong Identifiers"  = "black",
    "Weak Identifiers"    = "grey55",
    "Independent Leaners" = "black",
    "Pure Independents"   = "grey55")) +
  labs(
    x = "Year",
    y = "Proportion",
    title = "Figure 1: The Distribution of Party Identification, 2006-2024",
    subtitle = "Proportion of National Election Study Sample"
  ) +
  theme(
    legend.position.inside = c(0.18, 0.22),
    legend.title = element_blank(),
    legend.background = element_rect(fill = "white", color = "black",
                                     linewidth = 0.3),
    panel.grid.major = element_line(linetype = "dotted",
                                    color = "grey80", linewidth = 0.3),
    panel.grid.minor = element_blank(),
    axis.title.x = element_blank(),
    axis.title.y = element_blank(),
    plot.title = element_text(hjust = 0.5, size = rel(1.8), face = "bold"),
    plot.subtitle = element_text(hjust = 0.5, face = "bold"),
    panel.background = element_rect(fill = "gray99", color = NA),
    plot.background = element_rect(fill = "white", color = NA))
```
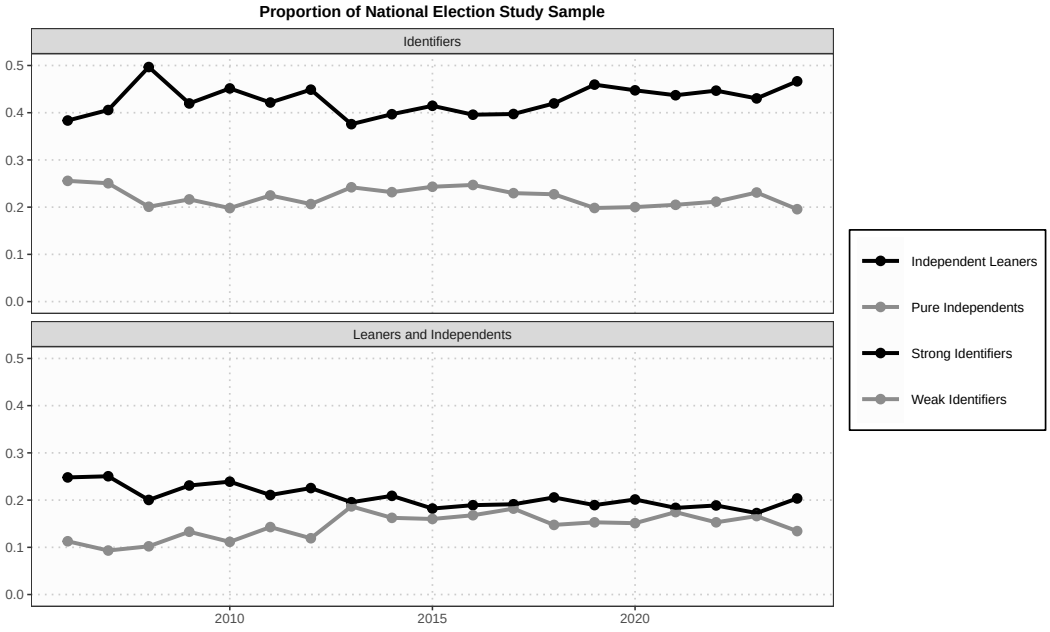
Figure 1: The Distribution of Party Identification, 2006−2024